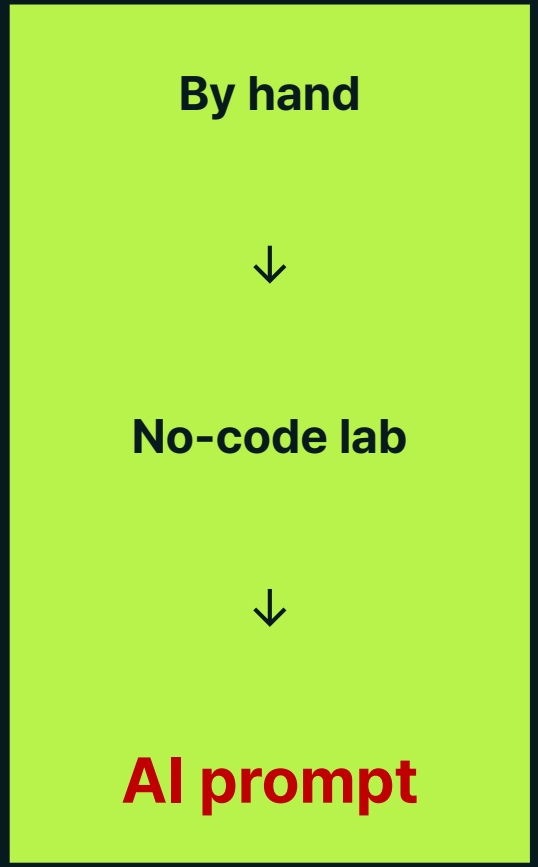
Reinforcement Learning Workbook

Step-by-step solutions you can follow with a calculator



Financial ML · Week 13 Workbook

At discount factor 0.9, investing beats taking now

Take now (10, 0, 0) · Invest (−2, 0, 20)

Step 1 · γ = 0.5

$$\gamma = 0.5: \quad -2 + 0.5(0) + 0.5^2(20) = -2 + 5 = 3$$

Step 2 · γ = 0.9

$$\gamma = 0.9: \quad -2 + 0.9(0) + 0.9^2(20) = -2 + 16.2 = 14.2$$

Step 3 · P(choose A) under ε-greedy (ε=0.2)

$$P(A) = 0.8 + \frac{0.2}{2} = 0.9, \qquad P(B) = 0.1$$

The learning rate changes how one experience updates Q

Q=2, reward 1, next max Q=4, γ=0.9

Step 1 · Update

$$Q(s,a) \leftarrow Q(s,a) + \underbrace{\alpha}_{\text{learning rate}} \left[\underbrace{r + \gamma \max_{a'} Q(s',a')}_{\text{new target}} - \underbrace{Q(s,a)}_{\text{old score}} \right]$$

Step 2 · Target and α=0.5 update

$$2 + 0.5\,(4.6 - 2) = 3.3, \qquad \text{new target} = 1 + 0.9 \times 4 = 4.6$$

Step 3 · α = 0.1, 0.5, 1

$$\alpha = 0.1: \; 2 + 0.1(2.6) = 2.26 \qquad \alpha = 0.5: \; 3.3 \qquad \alpha = 1: \; 4.6$$

With α=1, Q jumps to the target 4.6

Execution cost = impact cost + waiting cost

10,000 shares, i=0–9, source teaching model

Step 1 · All at once

All at once : $0.00002 \times 10000^2 = 2000,$ waiting 0

Step 2 · 1,000 × 10 times

1,000 shares, 10 times : $10 \times 0.00002 \times 1000^2 = 200,$ $0.01 \times 1000 \times (0 + 1 + \cdots + 9) = 450$

Step 3 · 2,000 × first 5

2,000 shares, first 5 : $5 \times 0.00002 \times 2000^2 = 400,$ $0.01 \times 2000 \times (0 + 1 + 2 + 3 + 4) = 200$

Zero drift removes waiting cost: 2,000, 200, 400

Check Q-learning and execution cost by calculator

Main deck numbers as is

1. Where

Use a calculator or AI chat.

2. Input

Enter the Q-learning example.

3. Output

α=0.5 gives an update of 3.3.

4. Check

Check α=0.1 gives 2.26 and α=1 gives 4.6.

Prompt for the AI chat

I am new to RL. Define state, action and reward for a large order. With Q=2, reward=1, next max Q=4, γ=0.9, tabulate updates for α=0.1, 0.5, 1. Explain the market impact vs. waiting cost trade-off, and separate fixed-schedule comparison from RL learning.

CALCULATE BY HAND.

CHECK THE DENOMINATOR.

THEN TRUST THE NUMBER.

Calculate once by hand
and the assumptions show.